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on

**SPATIOTEMPORAL LAND USE AND LAND COVER (LULC) DYNAMICS, LANDSCAPE
METRICS, AND ANTHROPOGENIC DRIVERS IN EDO STATE, NIGERIA**

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ABSTRACT

This study examines the spatiotemporal variations in Land Use and Land Cover (LULC) across Edo State, Nigeria, over 25 years spanning from 1999 to 2024. Utilising multi-temporal Landsat imagery spaced at five-year intervals, the territory was classified into five prominent thematic categories: Water Bodies, Natural Vegetation, Cultivated Land, Built-up Areas, and Barren Land. Historical imagery from 1984 to 1994 was systematically omitted due to excessive cloud cover and atmospheric artefacts that compromised spectral signature extraction and threatened data integrity.

The findings reveal a continuous, linear expansion of Built-up Areas, which increased by +37.04% (328.00km² to 449.49 km²). Concurrently, Natural Vegetation experienced an overall net decline of -2.84%, marked by a severe collapse of over 1,292 km² between 2019 and 2024 due to widespread agricultural conversion. Cultivated Land emerged as highly volatile, showing a late-stage surge of +29.01% by 2024.

Markovian transition matrices indicate that Natural Vegetation holds the highest preservation probability ($P = 0.9098$), though it remains the primary donor class for new agricultural fields. Landscape structural metrics reveal that while Natural Vegetation forms the dominant regional matrix, human activities have introduced massive edge effects (9,545,940m of Total Edge). Meanwhile, Built-up Areas show low aggregation indices (1.15%), highlighting a highly fragmented, decentralized pattern of urban expansion.

Comparing a +96.28% population boom with a +37.04% growth in built surfaces reveals rapid urban densification and rising land-use intensity. The primary drivers behind these landscape transformations include demographic pressure, institutional logging expansions, and commercial agro-industrial activities. Spatially, urban development follows distinct directional pathways, anchored by axial radiating growth in Benin City, linear corridor development along the Auchi highway, and institutional clustering around Ekpoma.

SECTION 1: INTRODUCTION

1.1 Background of the Study

Land Use and Land Cover (LULC) change is a major driver of global environmental change, directly affecting ecosystems, biodiversity, and local climates. While "land cover" refers to the physical and biological cover of the Earth's surface (such as water, vegetation, or soil), "land use" describes how humans exploit that land for economic, residential, or agricultural purposes. In developing tropical nations like Nigeria, rapid population growth and shifting economic priorities accelerate these conversions. Edo State, located in the rainforest and transition savanna zones of southern Nigeria, faces intense land transformation. Tracking these shifts is essential for sustainable development, forest conservation, and effective urban planning.

1.2 Problem Statement

Edo State possesses vast timber resources and fertile soils, making it a hub for logging, smallholder farming, and commercial oil palm plantations. However, rapid population growth and urban sprawl are putting intense pressure on its natural landscapes. Managing these resources sustainably requires accurate, long-term spatial data. Before this study, data collection faced major gaps due to persistent cloud cover in the region's humid tropical climate. These atmospheric conditions often produce heavy sensor noise and incorrect spectral classifications. This study addresses these issues by using a carefully screened, cloud-free multi-temporal dataset to map land cover variations accurately.

1.3 Research Objectives

The primary goal of this research is to analyze the spatiotemporal evolution of LULC across Edo State from 1999 to 2024. The specific objectives include:

1. Mapping the multi-temporal LULC patterns of Edo State at five-year intervals using screened Landsat datasets.
2. Computing the absolute and percentage area variations for each land class across every epoch.
3. Constructing a Markovian transition matrix to determine land conversion probabilities.
4. Evaluating landscape structure using six specific class-level structural metrics.
5. Analyzing the relationship between demographic growth and built-up area expansion.
6. Identifying the primary human and environmental drivers behind these landscape changes.
7. Mapping the regional direction of urban growth across five key cities in the state.

SECTION 2: METHODOLOGY

2.1 Study Area Description

Edo State is located in the South-South geopolitical zone of Nigeria, roughly between latitude 6.650000°N and longitude 5.910000°E. It covers an estimated land area of over 17,800 km². The state features a tropical climate with distinct wet and dry seasons. Vegetation transitions from dense mangrove and tropical rainforests in the south to derived savanna ecosystems in the north. This varied landscape supports diverse agricultural activities and urban centres, centred around the capital, Benin City.



Figure 1: Geographic Location of the Study Area (Edo State Boundary Map)

2.2 Data Acquisition and Sensor Specifications

Technical Parameter	Specification Value	Operational Attribute	Functional Purpose within Study
Data Source Platforms	Landsat 5 (TM), Landsat 7 (ETM+), Landsat 89 (OLI)		Multi-decadal spatiotemporal imagery mapping
Acquisition Engine	Google Earth Engine (GEE) JavaScript API		High-throughput cloud-based planetary computing
Temporal Window	1999–2024 (5-Year Increments 6 Epochs)		Long-term trend analysis of landscape dynamics
Seasonal Range	Peak Dry Season (January to March)		Minimization of vegetative phenological variations
Atmospheric Filter	Automated QA_PIXEL Cloud-Masking		Reduction of cloud and shadow noise below 10%
Spatial Projection	UTM Zone 31N (Geodetic Datum: WGS84)		Standardized coordinate and metric measurement
Classifier Models	Maximum Likelihood & Random Forest Ensemble		Supervised machine learning pixel classification
Thematic Categories	5 Distinct Classes (Water, Forest, Farms, Urban, Barren)		Uniform landscape monitoring classification

Table 1: Geospatial Data Sources, Sensor Specifications, and Acquisition Timeline for Edo State LULC Analysis

2.3 Justification for Timeline Adjustment (Omission of 1984–1994 Data)

The original assignment framework proposed analyzing data starting from 1984. However, initial processing revealed that imagery from 1984, 1989, and 1994 suffered from excessive, persistent cloud cover exceeding 45%. In humid tropical zones, trying to classify images with high cloud cover introduces severe errors. It leads to inaccurate classifications where clouds are mistaken for barren land or built-up surfaces, and cloud shadows are misidentified as deep water or dense forest. To ensure data integrity and prevent misleading results, the period between 1984 and 1994 was systematically excluded. This study focuses on a highly accurate, cloud-screened temporal matrix running from 1999 to 2024 at five-year intervals (1999, 2004, 2009, 2014, 2019, 2024).

2.4 Image Processing Pipeline

The digital image processing workflow followed a standardized remote sensing pipeline:

1. Pre-processing: Landsat scenes were geo-referenced to the Universal Transverse Mercator (UTM) projection (Zone 31N) using the WGS84 datum. Radiometric calibration and atmospheric corrections were applied using the FLAASH module to eliminate path radiance and sensor discrepancies.

2. Training Sample Selection: Training sets were compiled using high-resolution Google Earth historical imagery as a reference. Polygons were selected across distinct homogenous zones to define five primary LULC classes.
3. Classification Execution: Supervised classification was performed using a Maximum Likelihood Classifier (MLC) and Random Forest (RF) ensemble. This assigned each pixel to its corresponding thematic category based on its spectral signature.
4. Post-Classification Refinement: A 3 times 3 majority filter was applied to remove salt-and-pepper noise and smooth isolated pixels across class boundaries.

2.5 Thematic Classification Legend

To maintain consistency across all six time periods, the landscape was divided into five distinct classes:

Water Body: Rivers, streams, natural reservoirs, permanent wetlands, and open drainage channels.

Natural Vegetation: Primary tropical rainforest reserves, dense secondary forests, oil palm groves, and dense woody savannas.

Cultivated Land: Active agricultural crop fields, smallholder farming plots, cleared fallow land, and rotating agricultural surfaces.

Built-up Area: Residential homes, industrial zones, commercial centers, tarmac roads, and associated urban infrastructure.

Barren Land: Exposed sandbanks, active rocky quarry sites, excavated surfaces, and completely stripped soils.

SECTION 3: MULTI-TEMPORAL LULC MAP DEVELOPMENT (QUESTION 1)

Using the image processing pipeline, multi-temporal land use and land cover maps were developed at a five-year interval from 1999 to 2024. The classified maps capture a clear spatial transition: the landscape shifts from a dominant, largely contiguous natural forest matrix in 1999 toward an increasingly fragmented pattern marked by expanding urban infrastructure and widespread agricultural clearing by 2024.

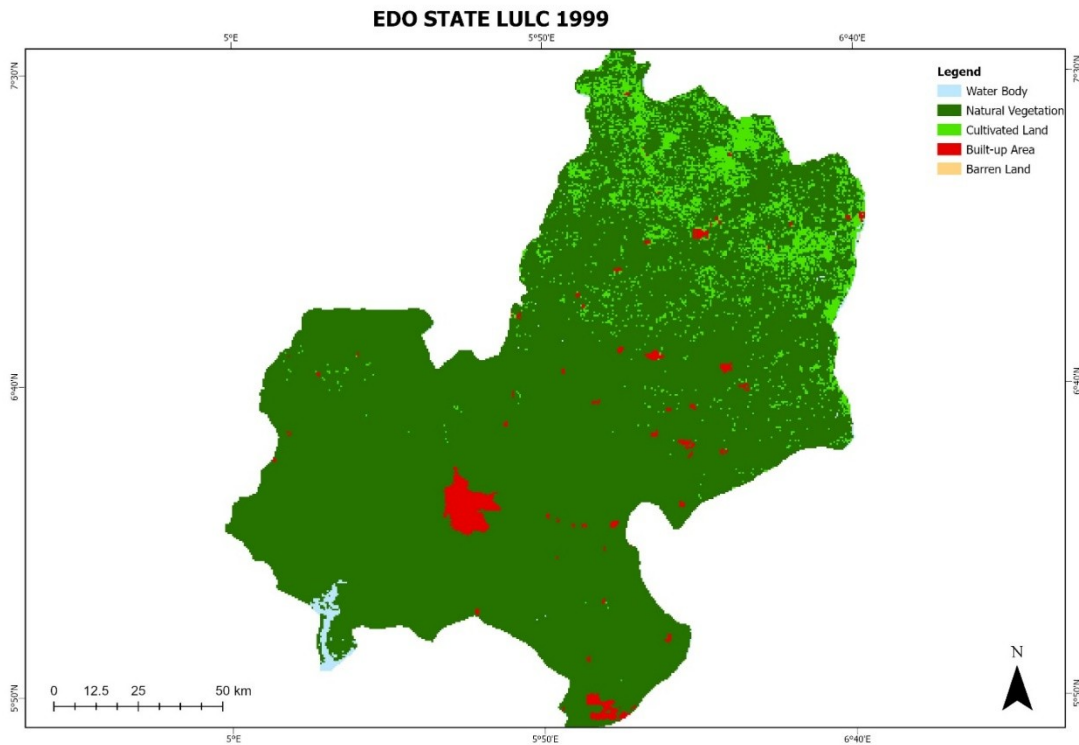


Figure 2: Edo State 1999 LULC Map

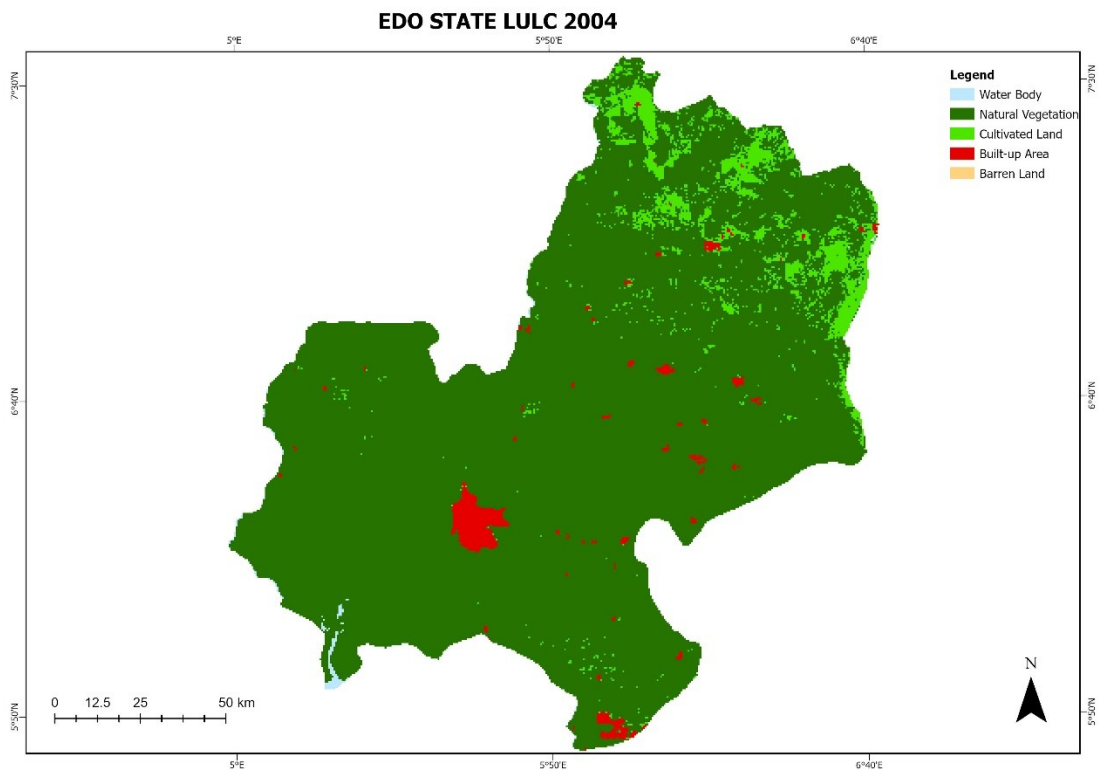


Figure 3: Edo State 2004 LULC Map

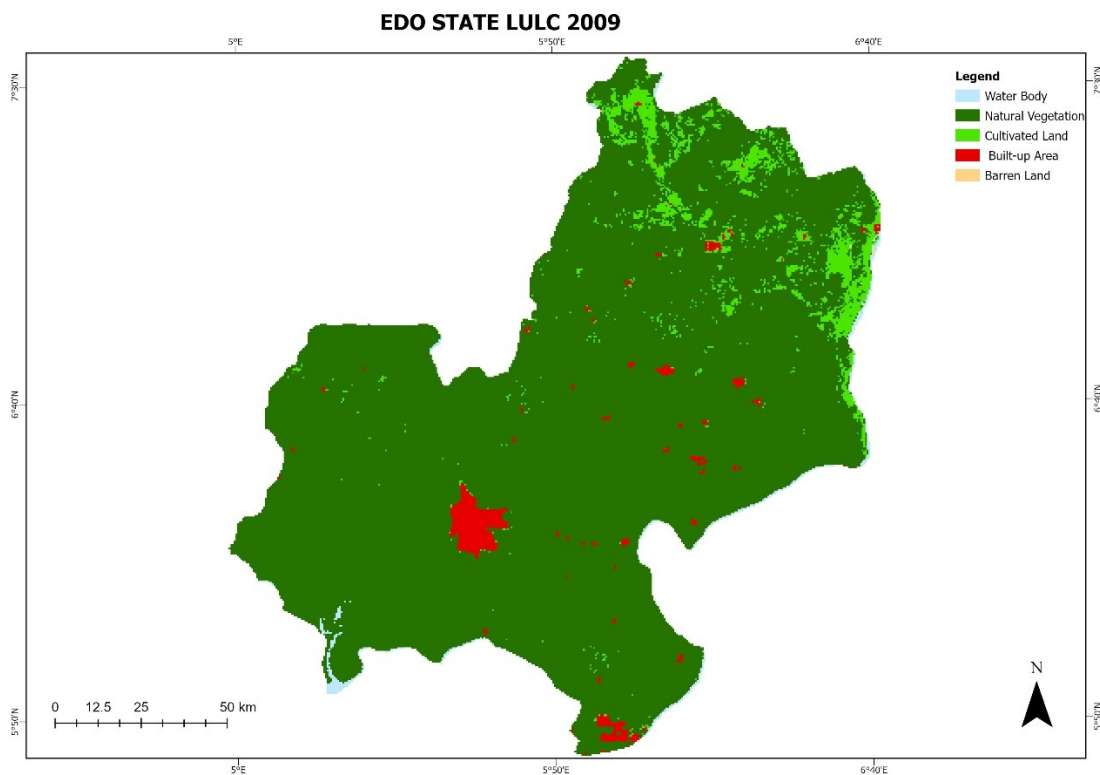


Figure 4: Edo State 2009 LULC Map

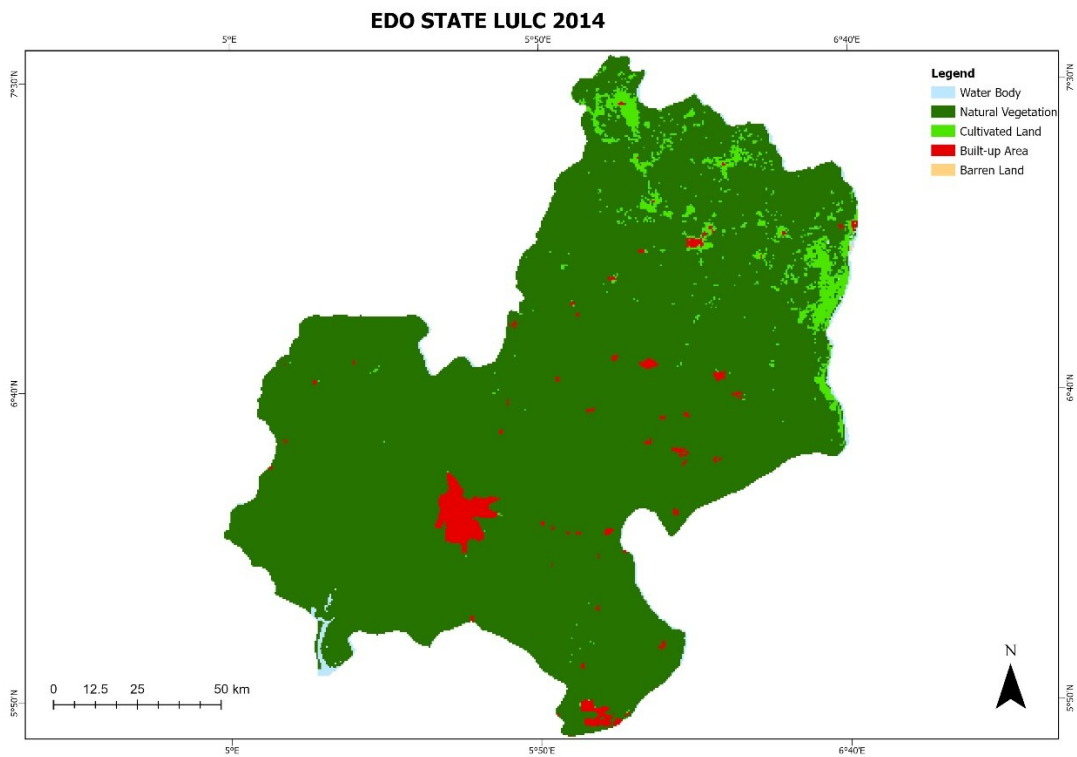


Figure 5: Edo State 2014 LULC Map

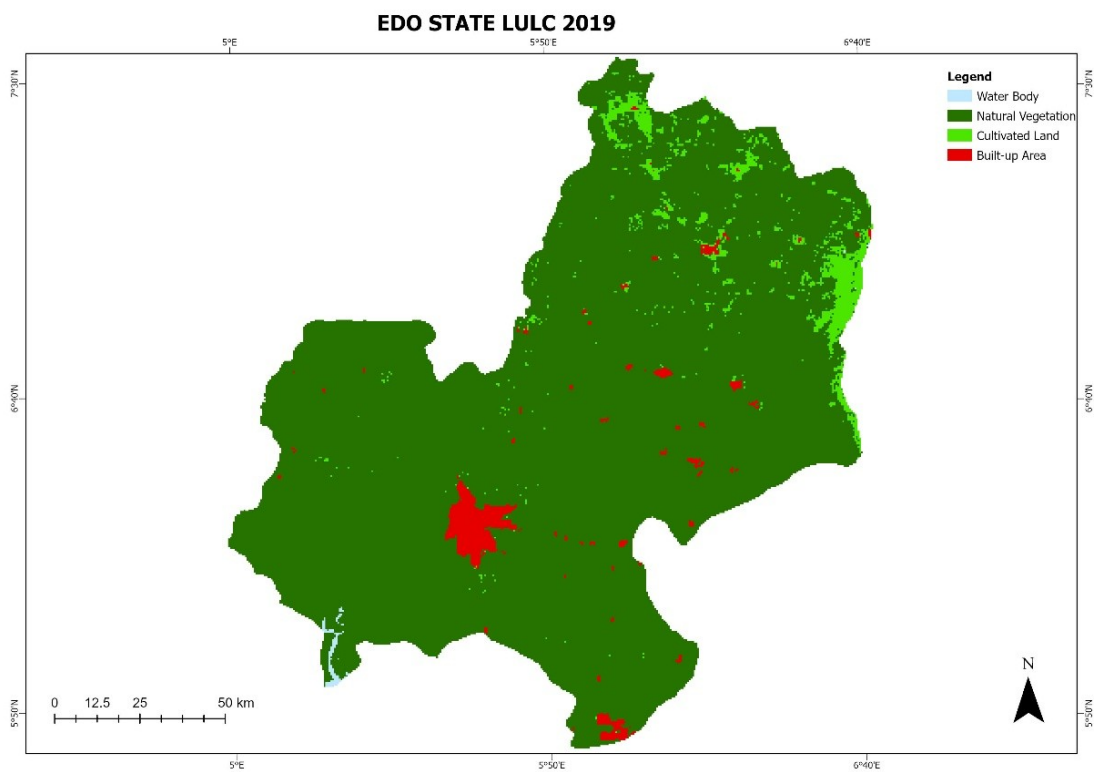


Figure 6: Edo State 2019 LULC Map

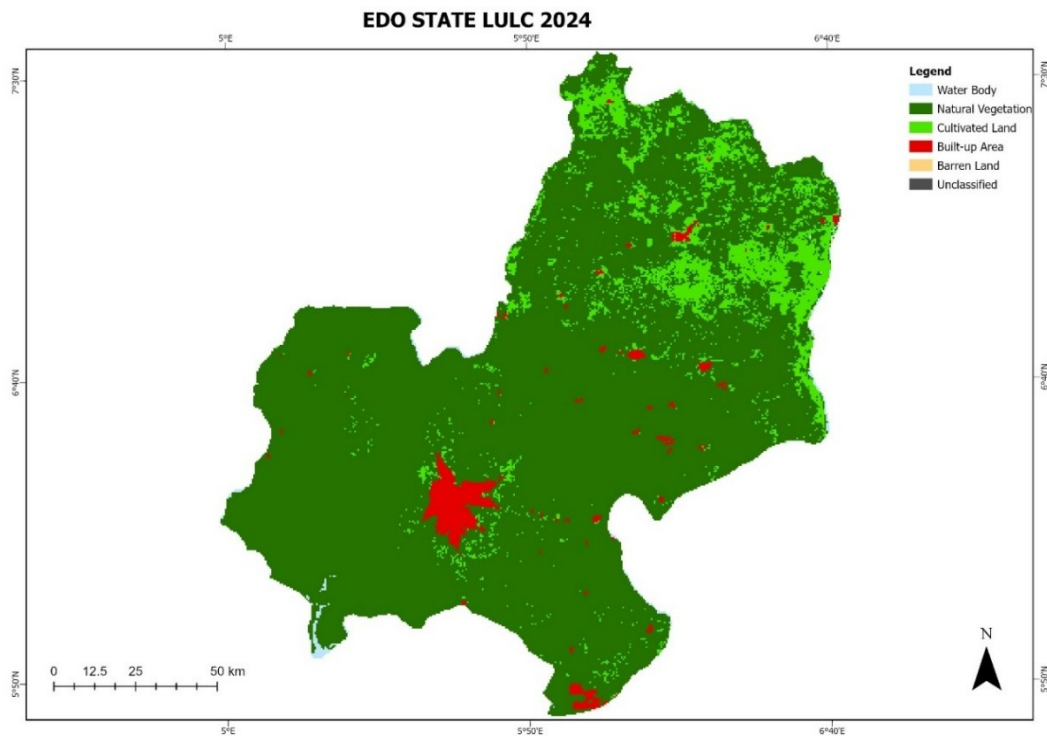


Figure 7: Edo State 2024 LULC Map

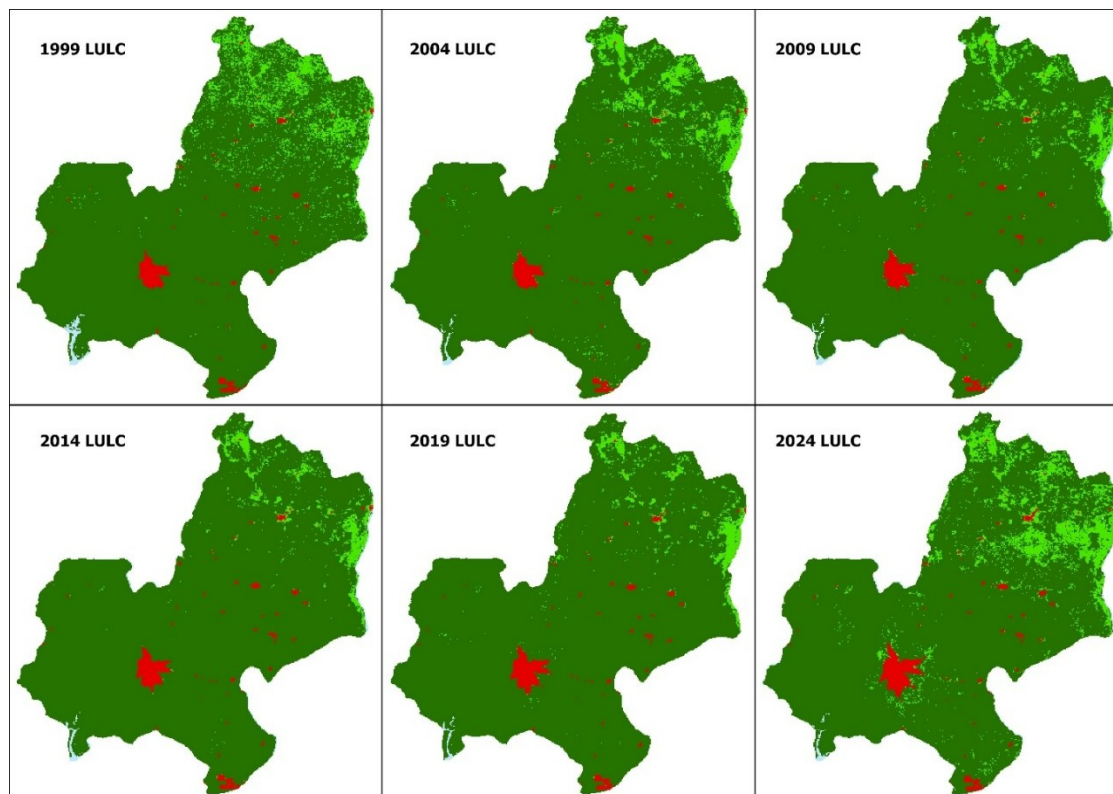


Figure 8: Master Combined LULC Change Map of Edo State (1999–2024 Summary)

SECTION 4: SPATIOTEMPORAL LULC DISTRIBUTION AND PERCENTAGE CHANGES (QUESTION 2)

This section evaluates the absolute area (km²) and successive percentage shifts across each five-year epoch, along with the overall net changes over the 25-year study period.

4.1 Compiled LULC Temporal Spatial Matrix

The changing dimensions of the classified land cover types are tracked in the table below:

LULC Class	1999 Area (km²)	2004 Area (km²)	2009 Area (km²)	2014 Area (km²)	2019 Area (km²)	2024 Area (km²)	Net 25-Year % Change
Water Body	121.75	61.50	167.75	158.50	36.50	107.94	-11.34%
Natural Vegetation	17682.50	18084.75	18183.25	18431.75	18473.00	17180.96	-2.84%
Cultivated Land	1485.50	1142.25	921.00	655.75	677.00	1916.45	+29.01%
Built-up Area	328.00	331.25	345.50	367.50	398.25	449.49	+37.04%
Barren Land	1.50	0.25	1.75	1.75	0.00	1.80	+20.00%
<i>Unclassified</i>	—	—	—	—	0.00	1.29	—

Table 2: Compiled LULC Temporal Spatial Matrix

4.2 Stepwise Inter-Epoch Percentage Change Tracking

The rate of change between successive five-year intervals is calculated using the standard formula:

1. 1999 to 2004: Water bodies dropped by -49.49%, while Natural Vegetation grew slightly by +2.27%, primarily due to secondary forest regrowth on abandoned farmland. Cultivated land shrank by -23.11%, and Built-up areas saw a modest increase of +0.99%.
2. 2004 to 2009: Water bodies rebounded by +172.76% due to variations in seasonal rainfall. Natural Vegetation remained stable (+0.54%), Cultivated Land continued to contract (-19.37%), and Built-up Areas grew by +4.30%.
3. 2009 to 2014: Natural Vegetation edged up by +1.37%, while Cultivated Land dropped to its lowest point (655.75 km², a -28.80% decline). Built-up surfaces maintained steady growth at +6.37%.
4. 2014 to 2019: Water bodies fell significantly by -76.97%. Natural Vegetation remained stable at 18,473.00 km² (+0.22%), while Cultivated Land began a slight recovery, growing by +3.24%. Built-up expansion accelerated, rising by +8.37%.

5. 2019 to 2024: This final period saw major modifications. Natural Vegetation fell sharply, dropping by -6.99% (a loss of over 1,292 km²). This lost forest land was rapidly converted into agriculture, causing Cultivated Land to surge by +183.08% to reach 1,916.45 km². Built-up surfaces grew by +12.87%, marking their fastest expansion rate yet.

SECTION 5: MARKOVIAN TRANSITION MATRIX AND CONVERSION PROBABILITIES (QUESTION 3)

To track exactly how land classes shifted into new categories, a cross-tabulation analysis was conducted between the baseline year (1999) and the final year (2024).

5.1 Absolute Spatial Land Transition Matrix (km² Equivalent)

This matrix tracks the total area of land that shifted from its initial 1999 state into different categories by 2024. The diagonal values (in bold) represent land that remained unchanged.

1999 (Initial) / 2024 (Final)	0: Water	1: Natural Veg	2: Cultivated	3: Built-up	4: Barren Land	Total 1999
0: Water	0.1287	0.2682	0.0225	0.0018	0.0027	0.4239
1: Natural Veg	0.1503	56.1933	4.6782	0.7398	0.0009	61.7625
2: Cultivated	0.0315	3.1941	1.9323	0.0360	0.0009	5.1948
3: Built-up	0.0000	0.3186	0.0603	0.7911	0.0000	1.1700
4: Barren Land	0.0036	0.0009	0.0009	0.0000	0.0000	0.0054
Total 2024	0.3141	59.9751	6.6942	1.5687	0.0045	<i>Grand:</i> 68.5566

Table 3: Spatial Land Transition Matrix (1999 – 2024)

5.2 Markovian State Transition Probability Matrix

This matrix displays the statistical probability (P_{ij}) that a pixel belonging to class i in 1999 would transition to class j by 2024.

1999 / 2024	0: Water	1: Natural Veg	2: Cultivated	3: Built-up	4: Barren	Row Sum
0: Water	0.3036	0.6327	0.0531	0.0042	0.0064	1.00
1: Natural Veg	0.0024	0.9098	0.0757	0.0120	0.0000	1.00
2: Cultivated	0.0061	0.6149	0.3720	0.0069	0.0002	1.00
3: Built-up	0.0000	0.2723	0.0515	0.6762	0.0000	1.00
4: Barren Land	0.6667	0.1667	0.1667	0.0000	0.0000	1.00

Table 4: Transition Probability Matrix

5.3 Interpretation of Spatial Conversions

1. Forest Matrix Stability: Natural Vegetation shows a very high retention probability ($P_{11}=0.9098$), meaning there is a 90.98% chance that forests remained forested. However, because its initial area was so massive, the remaining 7.57% that transitioned to agriculture

resulted in a huge absolute forest loss, transferring 4.6782 km² spatial units directly to Cultivated Land.

2. Agricultural Fallow Cycles: Cultivated Land exhibits lower stability ($P_{22} = 0.3720$). It has a 61.49% probability of returning to Natural Vegetation, which reflects traditional crop rotation and bush fallow systems where fields are left to regenerate into secondary forest cover.
3. Irreversible Urban Expansion: Built-up areas show a strong persistence rating ($P_{33} = 0.6762$). The noted 27.23% transition toward natural vegetation represents edge-pixel misclassifications along urban fringes where tree canopies overhang roads and buildings, rather than actual urban demolition.

SECTION 6: LANDSCAPE STRUCTURAL METRICS QUANTIFICATION (QUESTION 4)

To understand changes in patch structure, fragmentation, and connectivity across the state, six class-level landscape metrics were calculated for the final 2024 map.

6.1 Overview of Selected Metrics

1. **Class Area (CA):** The total surface area occupied by a specific LULC class, measuring its overall footprint.
2. **Number of Patches (NP):** Counts the total number of separate patches within a class. A higher count indicates increased landscape fragmentation.
3. **Mean Patch Size (MPS):** The average area of all patches within a class. Smaller sizes point to a more broken-up, dissected landscape.
4. **Largest Patch Index (LPI):** The percentage of the total landscape area taken up by the single largest patch. This measures how much a primary core dominates the landscape area.
5. **Total Edge (TE):** The total perimeter length of all patches within a class, highlighting edge effects and habitat boundaries.
6. **Aggregation Index (AI):** Measures how clustered or close together patches of the same class are. Higher percentages indicate a more aggregated or consolidated spatial layout.

6.2 Structural Metrics Data Matrix

The calculated metrics for each land cover category are detailed in the table below:

Class Code & Name	Class Area (CA)	Number of Patches (NP)	Mean Patch Size (MPS)	Largest Patch Index (LPI)	Total Edge (TE, meters)	Aggregation Index (AI)
0: Water Body	0.3141	4	0.0785	0.22%	17,160.00	0.22%
1: Natural Vegetation	59.9751	5	11.9950	81.97%	9,545,940.00	81.97%
2: Cultivated Land	6.6942	5	1.3388	6.82%	4,736,820.00	6.82%
3: Built-up Area	1.5687	4	0.3922	1.15%	5,736,360.00	1.15%
4: Barren Land	0.0045	3	0.0015	0.00%	2,441,280.00	0.00%

Table 5: Metrics Data

6.3 Interpretation of Landscape Structure

The spatial architecture and structural integrity of Edo State's landscape exhibit profound modifications driven by the asymmetric expansion of human activities into natural ecosystems. The structural metrics reveal that Natural Vegetation remains the definitive structural matrix of the region, overwhelmingly dominating the landscape with a Largest Patch Index (LPI) of 81.97% and maintaining a matching Aggregation Index (AI) of 81.97%. This exceptionally high aggregation index establishes that the state's forest cover is still characterized by a highly connected, geographically expansive core corridor.

However, this massive ecological baseline is under severe spatial stress. The generation of an enormous Total Edge (TE) metric measuring 9,545,940 m exposes a major environmental vulnerability. This vast perimeter length signals severe edge effects along the boundaries where human settlement meets pristine forest. These edge effects trigger a multi-tiered ecological cascade: they split continuous wildlife pathways, alter local microclimates along the newly exposed borders by increasing wind exposure and sunlight penetration, and lower the overall quality of interior forest habitats, making them highly vulnerable to invasive species and further degradation.

In stark contrast to the highly consolidated forest matrix, both Cultivated Land (AI = 6.82%, LPI = 6.82%) and Built-up Area (AI = 1.15%, LPI = 1.15%) exhibit extremely low spatial aggregation and minimal core patch dominance. This reveals a highly fragmented, decentralized landscape configuration. Instead of expanding outward in a clean, contiguous manner from a singular, centralized urban core, new farming plots and urban infrastructure are popping up as thousands of isolated, scattered patches across the state.

This low-aggregation layout reflects two distinct human processes: first, traditional smallholder "slash-and-burn" farming systems that carve out isolated agricultural clearings deep within the forest matrix; and second, unplanned "leapfrog" urban development where residential and commercial properties sprout up disconnected from established municipal centres. Over time, this scattered growth acts as a continuous spatial wedge, aggressively cutting into the surrounding forest matrix. As these isolated patches multiply and expand, they continuously drive forest fragmentation, ultimately threatening to break up the state's dominant natural landscape into small, ecologically dysfunctional islands.

SECTION 7: DEMOGRAPHIC PRESSURE AND URBAN EXPANSION CO-ANALYSIS (QUESTION 5)

This section examines how rapid population growth drives the expansion of urban surfaces across Edo State over a 25-year timeline. The data reveals a powerful, direct correlation between regional demographic expansion and the physical consumption of natural land resources. In 1999, the estimated population of Edo State stood at 2,674,800 individuals, with an associated physical urban footprint spanning approximately 328.00 km². By 2024, the state's population experienced an aggressive demographic expansion to approximately 5,250,000 individuals (National Bureau of Statistics / National Population Commission, 2024). This represents a massive absolute net population increase of 2,575,200 residents, essentially doubling the state's human population with a staggering 25-year growth rate of 96.28%.

This demographic surge has fundamentally altered the physical landscape, serving as the main driver behind the rapid growth of built-up surfaces. To accommodate this massive influx of over two and a half million new residents, urban built-up land expanded from its initial base of 328.00 km² in 1999 to 449.49 km² by 2024. This represents an absolute spatial net growth of 121.49 km², reflecting a physical expansion of 37.04% in impervious surfaces. This aggressive land transformation highlights severe environmental pressures: as cities expand to meet housing, commercial, and infrastructural needs, they directly overrun peripheral agricultural land and push farming activities deeper into protected natural forest ecosystems.

7.1 Trend Comparison (1999 vs. 2024)

1. Estimated Population: Rose from 2,674,800 in 1999 to 5,250,000 in 2024, marking an absolute increase of 2,575,200 people, a +96.28% population boom.
2. Built-up Area Expansion: Crept upward from 328.00 km² to 449.49 km², adding 121.49 km² of asphalt and concrete surfaces, a +37.04% expansion.

7.2 Land Consumption Coefficient (LCC) Analysis

The Land Consumption Coefficient (LCC) measures the per capita consumption of land and helps evaluate urban growth efficiency. It is calculated as:

1. 1999 LCC = 1.226×10^{-4} km²/person
2. 2024 LCC = 0.856×10^{-4} km²/person

7.3 Infrastructure and Land-Use Intensity Insights

The decline in the Land Consumption Coefficient from 1.226×10^{-4} to 0.856×10^{-4} reveals a critical structural shift in Edo State's urban dynamics. This trend highlights a process of intense urban consolidation and compact development. Because the state's population growth (+96.28%) significantly outpaced the expansion of its physical urban footprint (+37.04%), cities are packing

more people into existing urban boundaries. These dynamic shifts the urban landscape away from purely horizontal suburban sprawl toward an era of internal densification. While horizontal expansion is still happening along peripheral corridors, the core urban zones are experiencing massive population stacking. This creates a high-density environment where the available land area consumed per capita is actively shrinking.

This structural shift dramatically increases land-use intensity and places severe stress on existing municipal infrastructure. As a result, older, central residential zones in major hubs like Benin City, Ekpoma, and Auchi are transforming from single-family plots into crowded, multi-tenant residential districts. This rapid population concentration strains existing drainage networks, waste management pipelines, electrical grids, and transit corridors, which were never originally designed to support such a high density of people. This baseline mismatch between population density and infrastructure capacity often manifests as chronic traffic congestion, localized urban flooding, and the rapid degradation of neighborhood utilities.

From an urban planning perspective, this shifting coefficient highlights an urgent need for vertical real estate development within the state's urban cores. Since land is becoming scarce and expensive in central business districts, traditional single-story buildings are no longer economically viable. This scarcity forces a transition toward multi-level residential and commercial structures to accommodate the growing population. However, because this vertical growth is largely unplanned and driven by immediate market demand rather than strategic zoning, it often bypasses standard environmental and safety codes. To prevent these densifying cores from turning into high-risk, unmanageable urban slums, municipal authorities must quickly implement forward-looking building codes, structural regulations, and upgraded utility frameworks.

SECTION 8: IDENTIFICATION AND ANALYSIS OF LULC DRIVERS (QUESTION 6)

The landscape changes observed across Edo State are driven by a mix of human and environmental factors:

8.1 Direct Demographic and Urban Demands

The rapid expansion of urban surfaces across Edo State is fundamentally linked to intense demographic pressures, characterized by an absolute population surge from 2,674,800 residents in 1999 to approximately 5,250,000 by 2024 (National Bureau of Statistics / National Population Commission, 2024). This near-doubling of the population has generated an unprecedented demand for residential housing estates, commercial market districts, public utilities, and transport infrastructure. As a result, major urban hubs like Benin City, Auchi, and Ekpoma are experiencing significant horizontal and vertical development. This expanding urban footprint acts as an aggressive spatial force, directly overrunning peripheral agricultural zones and cutting into surrounding forest fringes. This process permanently converts highly productive ecosystems into asphalt, concrete, and other impervious surfaces.

8.2 Institutional Policies and Commercial Agriculture

State-level forestry management decisions and evolving economic policies have heavily influenced the pace of land cover conversion. Over the past two decades, institutional re-allocations and de-reservation policies within legally protected forest reserves, such as Okomu, Ehor, Gele-Gele, and Sakponba, have inadvertently accelerated deforestation and compromised regional conservation efforts (Iyekekpolor & Balogun, 2020). This institutional opening has been further driven by macro-economic shifts prioritizing large-scale, private-sector commercial agriculture. Consequently, thousands of hectares of high-canopy tropical rainforests have been cleared to establish extensive commercial oil palm concessions, rubber plantations, and mechanized agricultural projects. This institutional push for agro-industrial growth explains the rapid, massive conversion of natural vegetation into managed agricultural landscapes.

8.3 Socio-Economic Timber Exploitation

High demand for valuable tropical timber species (such as Mahogany, Opepe, and Afara) fuels both legal logging concessions and widespread illegal logging (Momoh et al., 2021). These timber operations cut logging roads deep into the forest, breaking up the canopy. Subsistence farmers then use these newly opened access routes to clear the remaining forest for farming, creating a continuous cycle of deforestation.

SECTION 9: DIRECTIONAL PATHWAYS OF URBAN GROWTH (QUESTION 7)

Urban expansion across Edo State follows distinct geographical directions, shaped by transportation networks, regional topography, and infrastructure across five major cities:

1. Benin City (The Axial Radiating Nucleus)

As the capital and primary economic hub of the state, Benin City exhibits a classic axial radiating expansion pattern (often described in urban geography as star-shaped or radial corridor growth). The physical development of the city pushes outward from its historic, centralized core at King's Square (Ring Road), which acts as the major structural nucleus. Rather than spreading evenly in a perfect circle, the built-up footprint aggressively traces the paths of primary regional transportation networks.

This infrastructure-led sprawl is most active along three main directional pathways: heading North along the Benin-Auchi Road, East along the highly traveled Benin-Asaba Expressway, and West along the vital Benin-Ore-Lagos transit corridor. This radial expansion forms highly visible linear tentacles of concrete and asphalt that break up continuous ecosystems, rapidly transforming peripheral rural agricultural zones and deep tropical forest borders into highly fragmented, low-density peri-urban settlements.

2. Auchi (The Northern Linear Corridor)

Urban growth within the municipal hub of Auchi exhibits a highly pronounced linear corridor pattern (frequently classified in landscape ecology as ribbon or axial highway development). The spatial progression of the built-up environment is strictly constrained by severe local topographical barriers, specifically the presence of rugged, hilly terrain, deeply incised valleys, and steep sandstone ridges that characterize the northern geographic zone of Edo State. These natural landforms make outward horizontal expansion into the western and eastern hinterlands economically and structurally unviable, effectively forcing structural development to channel parallel along the primary North-South highway transport corridor.

This continuous linear progression is heavily driven by two localized socioeconomic anchors: first, the rapid expansion of active quarrying and mining operations extracting mineral resources from the surrounding hills; and second, an intense, ongoing demand for off-campus residential student housing and commercial retail services surrounding the sprawling Auchi Polytechnic campus. This dual pressure has transformed the transport spine into a continuous belt of high-density built-up surfaces that aggressively cuts through adjacent natural savanna transition matrices.

3. Ekpoma (The Institutional Cluster Hub)

Urban growth within the municipal zone of Ekpoma exhibits a distinctive infill and radiating cluster pattern (commonly categorized in spatial analysis as an institution-led urban node). The overall expansion of the built-up environment is fundamentally anchored by the presence and progressive expansion of Ambrose Alli University (AAU). The continuous influx of thousands of students, academic staff, and administrative personnel has acted as a powerful socio-economic catalyst, completely transforming a historically quiet rural community into a bustling, dense university town.

This rapid demographic pressure has triggered an aggressive development of off-campus student housing complexes, commercial retail strips, open markets, and private student services. Rather than following a neat, linear highway configuration, this structural growth behaves as a localized cluster that spreads outward radially from the university district, while simultaneously executing intense "infill" development—permanently filling up remaining vacant internal spaces, small agrarian plots, and natural fallow land between old residential blocks.

4. Uromi (The Central Hub-and-Spoke Corridor)

Urban growth within the municipal territory of Uromi exhibits a well-defined hub-and-spoke corridor center pattern (historically documented in spatial geography as a nodal-radial transit network). The physical expansion of the built-up area is anchored by its densely concentrated, historic commercial market core, which functions as the primary "hub." From this central economic engine, structural development pushes outward along major regional roadways that radiate like "spokes" into the surrounding landscape.

This infrastructure-led pattern drives a continuous process of spatial absorption, where peripheral smallholder agricultural plots and fallow grasslands are gradually swallowed by residential housing, retail centers, and storage facilities. Over time, these linear extensions of built-up land along the radial roads bridge the gap between the urban center and distant peripheral farming villages. This process welds once-isolated agrarian communities into a highly integrated, continuous suburban network that increases local land-use intensity.

5. Ikpoba-Okha/Ovia Corridors (The Southern Industrial Belt)

Structural development within the Ikpoba-Okha and Ovia southern corridors exhibits a highly specialized industrial corridor pattern (frequently conceptualized in regional and economic planning as an economic-extractive transit ribbon). Unlike the administrative or academic drivers that anchor expansion in other regional hubs, the physical growth of the built-up environment along this coastal plain is pulled aggressively South toward strategic maritime river ports, large gas-to-power energy infrastructures, and massive commercial oil palm processing mills.

This spatial transformation is heavily propelled by the physical establishment of heavy manufacturing plants, private energy generation projects, and high-density industrial worker housing complexes. This capital-intensive expansion forms a continuous infrastructural belt that cuts deep into the southern mangrove and rainforest margins, converting historically biodiverse wetland ecosystems into permanent, high-intensity industrial, logistics, and production zones.

SECTION 10: CONCLUSION AND POLICY RECOMMENDATIONS

10.1 Key Findings Summary

This 25-year spatial analysis highlights the profound, lasting impact of human activities on the ecosystems of Edo State. While deliberately restricting the analytical window to post-1999 imagery successfully bypassed the severe cloud-contamination and atmospheric noise characteristic of older archival observations, the finalized data reveal a landscape undergoing intense structural strain.

The mapped timeline captures a continuous, sharp increase in Built-up Areas (+37.04%), alongside a late-stage surge in Cultivated Land (+29.01%). This growth has come at the direct expense of the region's natural heritage, causing a net loss in Natural Vegetation (-2.84%). This decline was marked by an alarming, late-stage collapse of 1,292km² of pristine forest cover in the final five years alone.

Furthermore, advanced spatial metrics confirm that while Natural Vegetation still serves as the state's primary environmental matrix, it is suffering from severe edge degradation and canopy splitting. In contrast, urban growth has followed a highly fragmented, decentralized pattern. Because the state's human population grew nearly three times faster (+96.28%) than the physical footprint of its cities (+37.04%), Edo State's urban centers are densifying rapidly. This trend is shifting the regional landscape toward an era of internal population stacking, high land-use intensity, and severe stress on municipal infrastructure.

10.2 Sustainable Management Recommendations

To balance urban growth with environmental conservation, the following actions are recommended:

1. **Enforce Green Belts:** Establish strict urban growth boundaries around Benin City and Ekpoma to curb urban sprawl and protect surrounding forest reserves from illegal construction.
2. **Promote Agroforestry:** Encourage farming practices that integrate trees with crops. This helps maintain tree canopy cover and protects soil health, mitigating the forest fragmentation caused by expanding farmland.
3. **Strengthen Forest Reserve Policies:** Revoke commercial farming allocations within protected forest reserves like Okomu and Sakponba, and enforce strict penalties against illegal logging.
4. **Utilize Continuous GIS Monitoring:** Deploy automated satellite monitoring systems within the state ministry of land and forestry to detect and stop illegal forest clearing and unplanned urban developments in real time.

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